

Multiagent Infectious Disease Propagation & Outbreak Prediction

Integrating Agent-Based Modeling, Epidemiological Theory, and ML Forecasting

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Problem Statement: The Dynamic Challenge of Disease Prediction

Understanding and accurately predicting the spread of infectious diseases is paramount for public health. Traditional methods often lack the granularity to capture complex human interactions or the predictive power to forecast outbreaks in real-time. There is a critical need to combine dynamic simulations, classical epidemiological models, and data-driven approaches for comprehensive insights.

Complexity of Spread

Disease propagation involves intricate human behaviour and environmental factors.

Need for Integration

Combining micro-level simulations with macro-level models for robust predictions.

Timely Intervention

Accurate forecasts enable proactive public health interventions and policy planning.

Project Overview: A Hybrid Modeling System

Our project integrates three distinct methodologies to create a robust system for simulating and predicting disease outbreaks. This hybrid approach leverages the strengths of each model, building a comprehensive understanding of disease dynamics based on real COVID-19 data.



Multi-Agent Simulation (MAS)

Micro-level interactions and individual-based disease propagation.



Classical SIR Model

Macro-level epidemiological dynamics and theoretical understanding.



Machine Learning (ML) Predictions

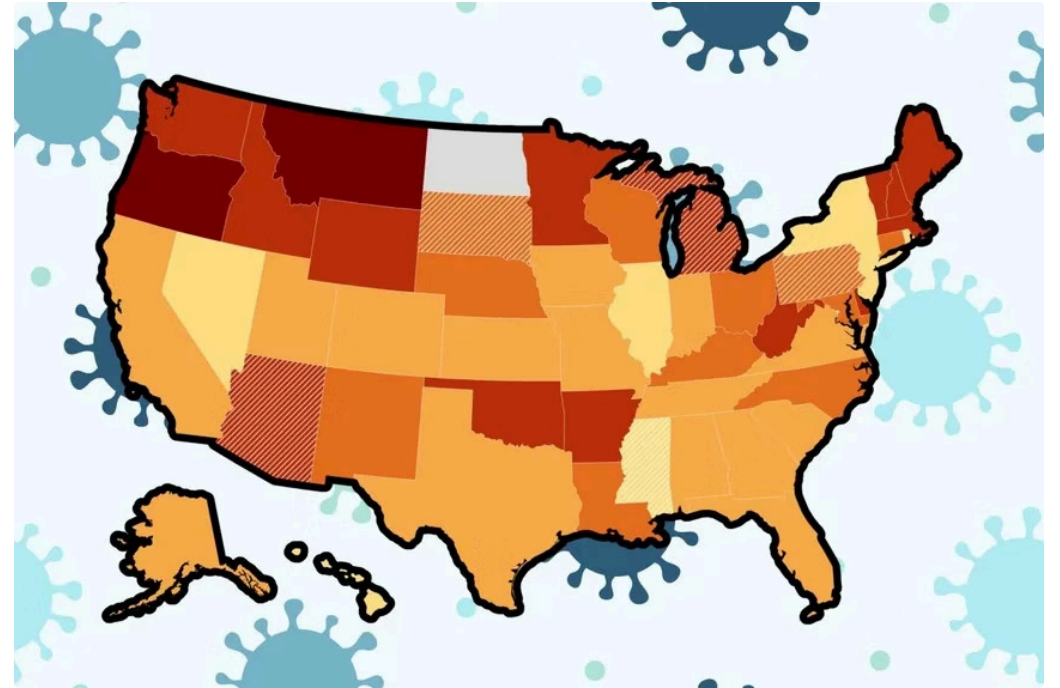
Data-driven forecasting and pattern recognition for outbreak prediction.

The models are grounded in real-world COVID-19 data, ensuring practical applicability.

Dataset: COVID-19 Case Data

Our analysis leverages a comprehensive COVID-19 dataset, providing a granular view of global infection trends. This dataset spans 198 countries, encompassing 161,568 records, offering a rich source for epidemiological modelling.

- **Source:** COVID-19 dataset from datasets/covid-19
- **Global Coverage:** 198 countries
- **Total Records:** 1,615,680



Focused Analysis: For this project, we specifically analysed the US data, comprising 816 records from January 2020 to April 2022. This focused approach allowed for in-depth model training and validation.

Multi-Agent Simulation (MAS): Micro-level Dynamics

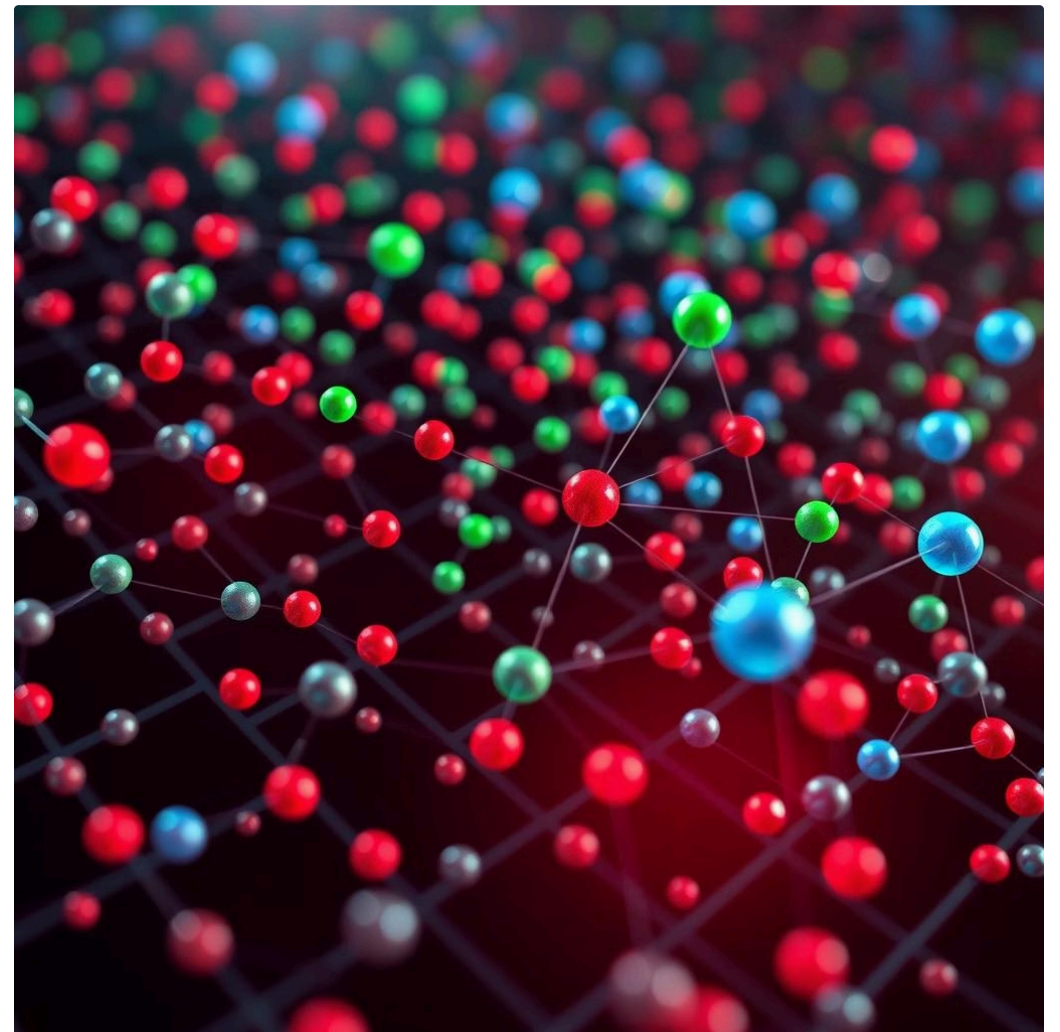
Our Multi-Agent Simulation models disease spread by tracking individual agents' states and interactions. This bottom-up approach allows for capturing complex, emergent behaviours that macro-level models might miss.

Agent Characteristics

- **Population:** 1000 agents
- **Mobility:** Random, simulating daily movements
- **Demographics:** Age distribution included

Disease States

- **Susceptible (S):** Healthy individuals
- **Infected (I):** Actively spreading the disease
- **Recovered (R):** Immune or no longer infectious
- **Dead (D):** Deceased due to infection



Transmission Mechanism

Proximity-based transmission modelled via a dynamic contact network, where agents within a defined radius can infect susceptible individuals.

Multi-Agent Simulation Results: Early Spread Dynamics

Initial MAS runs demonstrated the progression of infection under specific parameter settings. The simulation provides a granular view of how the disease spreads from a single infected individual within a mobile population.

Step 0	999	1	0	0
Step 20	993	2	5	0
Step 40	989	2	9	0

 **Observation:** The initial spread was limited. This indicates the sensitivity of the MAS model to parameters such as agent mobility, infection rate, and contact radius. Fine-tuning these parameters is crucial for realistic simulations.

Classical SIR Model: Macro-level Epidemiology

The Susceptible-Infected-Recovered (SIR) model provides a foundational understanding of epidemic dynamics. Using a system of ordinary differential equations (ODEs), it simulates how a disease progresses through a population at a macro level.

Key Parameters

- **Infection Rate (β):** 0.3 (average number of contacts per infected individual per day)
- **Recovery Rate (γ):** 0.1 (proportion of infected individuals recovering per day)

Basic Reproduction Number (R_0)

Calculated as $R_0 = \beta / \gamma = 0.3 / 0.1 = 3.0$.

This indicates that, on average, each infected person will infect 3 other susceptible people in a fully susceptible population, suggesting a significant potential for epidemic growth.

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta IS}{N} \\ \frac{dI}{dt} &= \frac{\beta IS}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

These equations describe the rate of change for each population compartment: Susceptible (S), Infected (I), and Recovered (R).

Machine Learning Models: Outbreak Prediction

To complement the simulation and theoretical models, we employed machine learning techniques for data-driven outbreak prediction. These models learn patterns from historical data to forecast future trends in cumulative confirmed cases.



Random Forest Regressor

An ensemble method that combines multiple decision trees for robust predictions, effectively handling non-linear relationships in the data.



Linear Regression

A fundamental statistical model for understanding the linear relationship between variables, providing a baseline for predictive performance.

Training Data: Both models were trained on the cumulative confirmed cases from the US COVID-19 dataset. A windowed feature extraction approach (lag-based) was used to capture temporal dependencies in the data.

ML Results: Performance Metrics

The evaluation of our machine learning models reveals strong performance in predicting cumulative confirmed cases. Both Random Forest Regressor and Linear Regression achieved high R^2 values, indicating a good fit to the historical data.

MAE (Mean Absolute Error)	57966.77	35818.76
RMSE (Root Mean Squared Error)	106262.83	88136.42
R^2 (Coefficient of Determination)	0.91	0.93

The near-perfect R^2 score suggests that both models captured the variance in the training data effectively, with Linear Regression showing slightly better MAE and RMSE, indicating closer predictions on average.

Prediction Evaluation: Test Set Performance

Evaluating the models on an unseen test set provides a realistic assessment of their generalization capabilities. The results demonstrate the strong predictive power of our models for forecasting disease outbreaks.

112,644...

Test MAE

Average absolute difference
between predicted and
actual values.

130,42...

RMSE

Measures the square root of
the average of the squared
errors.

0.7758

R^2

Proportion of variance in the
dependent variable
predictable from the
independent variables.

0.9952

Pearson Correlation

Strong linear relationship
between predicted and
actual values.